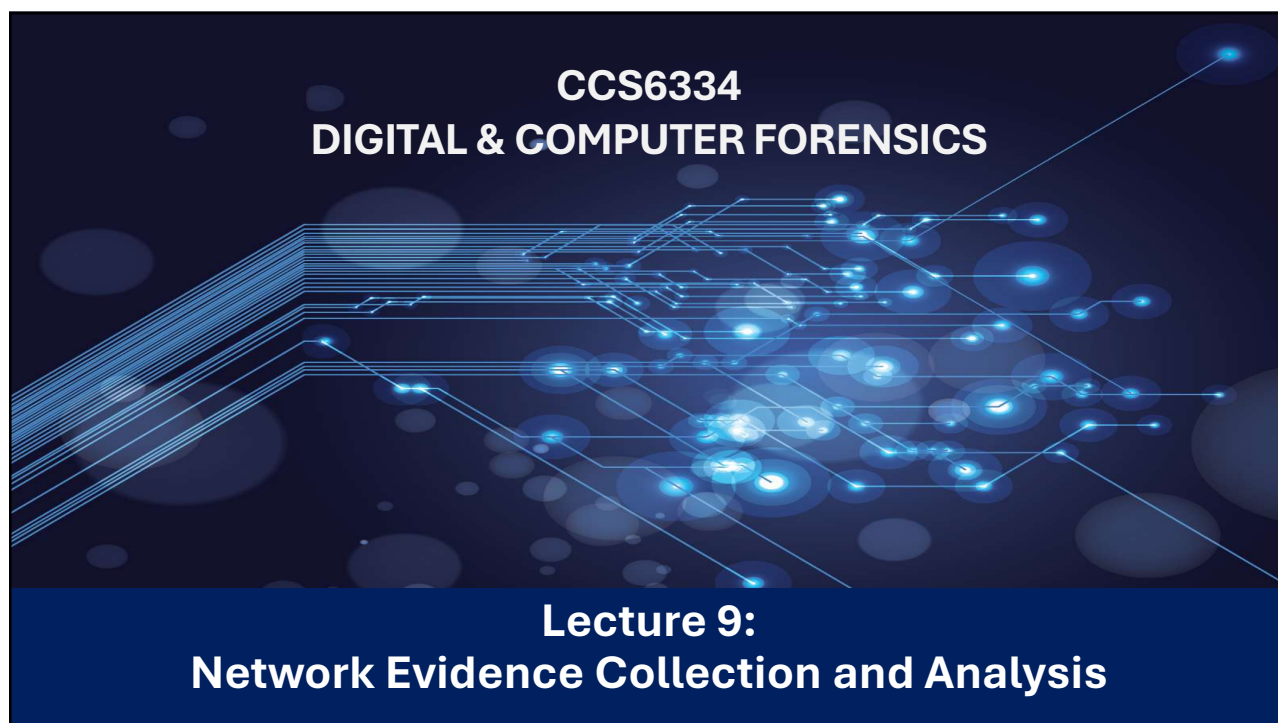




1

Learning Objectives

After studying this chapter, you should be able to:

- To examine and analyze network device evidence
- To capture and analyze network traffic
- To analyze network log files



2

NETWORK EVIDENCES



3

Using Log Files as Evidence



- Log files contain primary records of a person's activities on a system or network
- Log files can often identify:
 - Source, nature, and time of an attack
 - Specific user account of events related to illicit activities

4



Types of Logs

Authentication

Application

Operating system event

Network device

Security event



5

NETWORK DEVICE EVIDENCES


Identification


Preservation


Analysis


Documentation


Presentation



6



Router Forensics

- Router is hardware or software device that forwards data packets across a network to a destination network
 - May contain:
 - Read-only memory (ROM) with power-on self test code
 - Flash memory containing the router's operating system
 - Non-volatile random access memory (RAM) containing configuration information
 - Volatile RAM containing routing tables and log information
- Getting Evidence from the Router**
- Do not shut down router
 - Can erase valuable evidence
 - Do not alter anything
 - Document your process
 - Connect with the router to run certain commands

7



Switches Forensics

Two key points of evidence

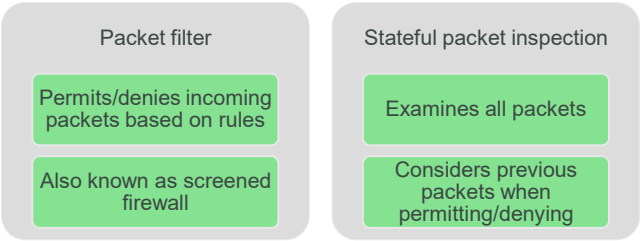
- Content Addressable Memory (CAM) table that maps the physical ports on the switch to the NIC on each device connected to the switch.
 - Incident responders tracing connections to specific network jacks can utilize this information which can aid the identification of possible rogue devices such as wireless access points or systems connected to the internal network by an adversary → often found via syslogs
- The second way in which switches can aid an incident investigation is through facilitating network traffic capture – either through the devices themselves (device logging/redirection) or through wire-tapping

8



Firewall Forensics

- Many current firewalls contain a wide variety of features such as intrusion detection and prevention, web filtering, data loss prevention, and detailed logs about allowed and denied traffic
- Often, firewalls serve as a detection mechanism that alerts security personnel to potential incidents → This can include alerts from features such as IDS/IPS systems, blacklists, or alerts flagging configuration changes to the firewall without the knowledge of IT personnel.
- Incident responders should have as much visibility of how their organization's firewalls function and what data can be obtained prior to an incident
- Acquiring evidence from firewalls is largely dependent on the manufacturer and the specific model that is used and the output are usually connection logs and access logs

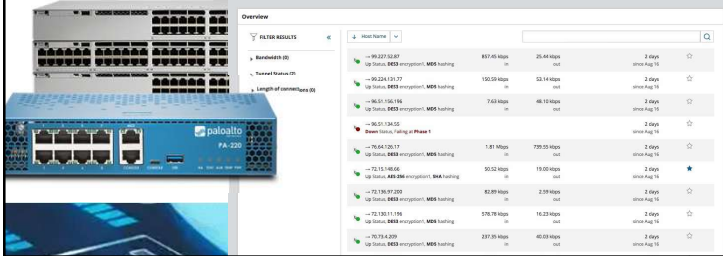


9



Collecting Data from Firewalls

- All the traffic going through a firewall is part of a connection.
- A connection consists of two IP addresses communicating with each other and two port numbers that identify the protocol or service
- Attempts on same set of ports from many different internet sources are usually due to decoy scans
- Carefully check firewall logs for any sort of connections or attempted connections on those ports
- Use protocol analysis to determine who attacker is



Type of Port	Port Numbers
Well-known ports	0 to 1023
Registered ports	1024 to 49151
Dynamic ports	49152 to 65535

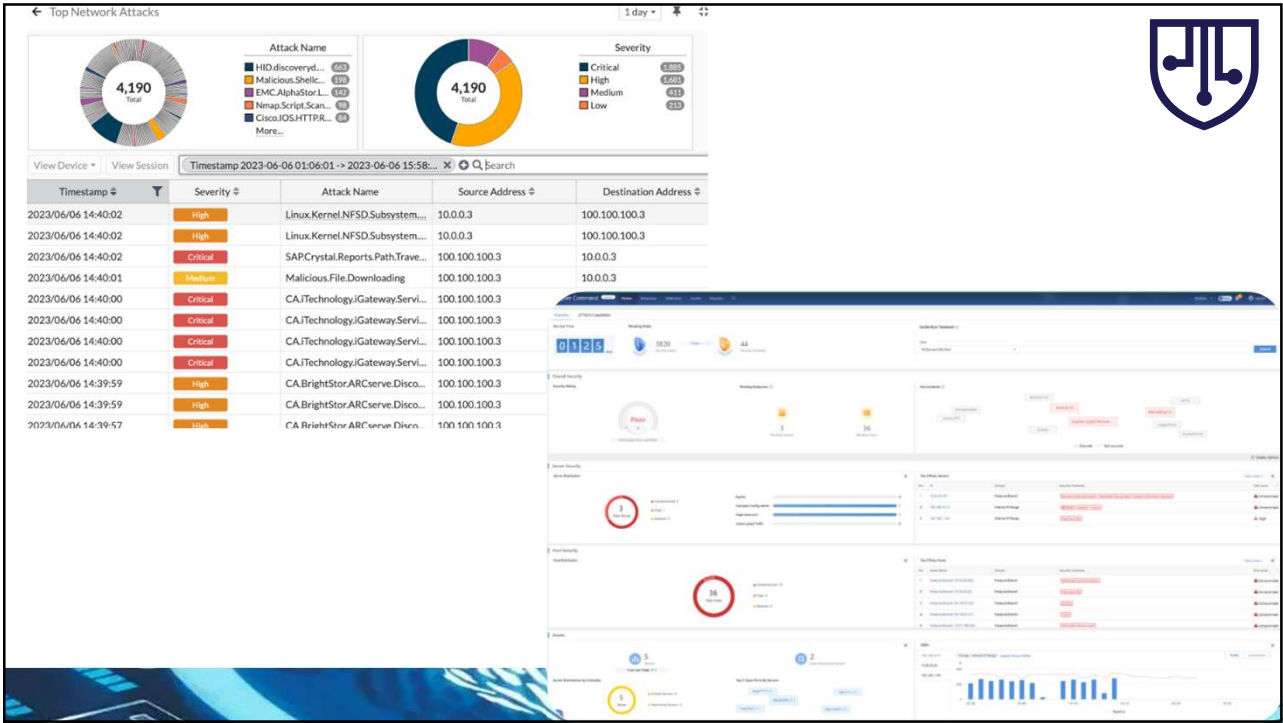
10



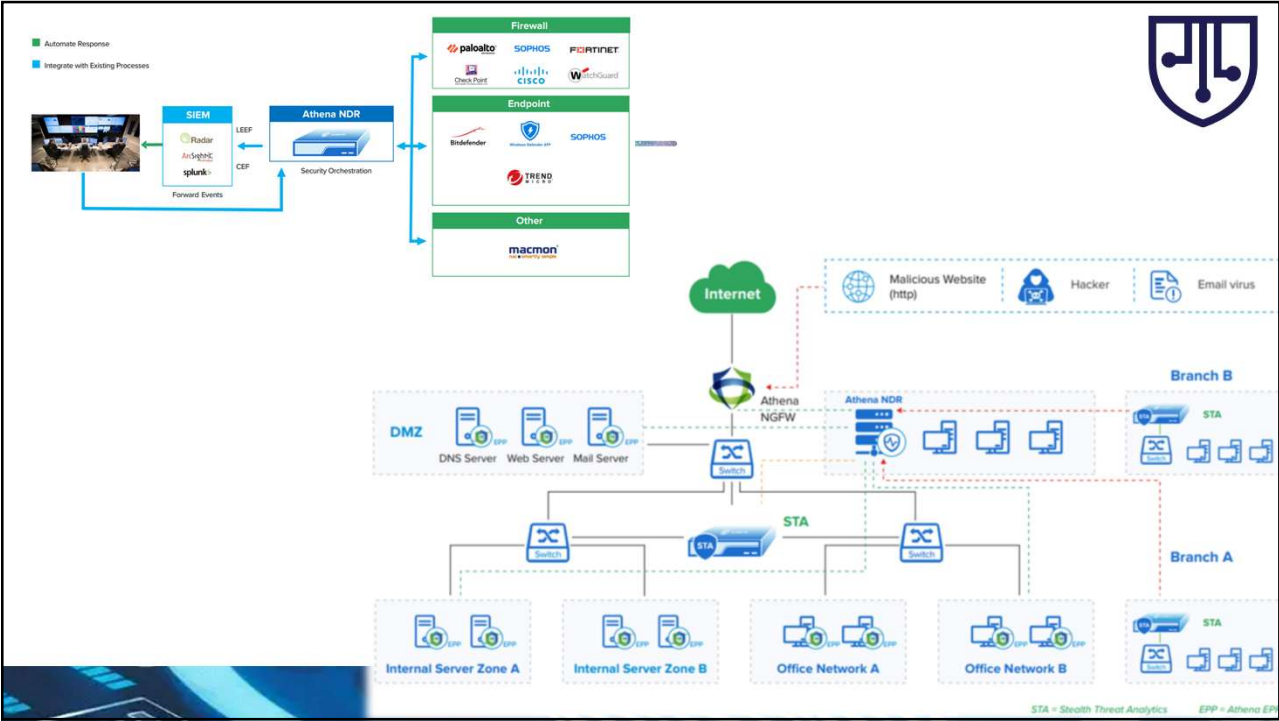
Network based IDS/IPS/DR

- NIDS/NIPS Sometimes found built into firewalls or as standalone network devices (appliance-based)
- Often configured to alert you to a specific malicious activity while an Intrusion Prevention System (IPS) can detect, but also block, potential malicious activity.
- In either case, both types of platform log are an excellent place for incident responders to locate specific evidence on malicious activity.
- NDR (Network Detection & Response): Comprehensive, deep network visibility, with AI/Machine Learning + Behavioral Analytics
 - Action: Detects, investigates, provides context, and can trigger automated responses (via integrations).

11



12



13

Other Source of Network Evidences

DHCP SERVERS	DOMAIN CONTROLLERS
• Standalone or built-in to other devices, DHCP servers manage a list of assigned IP addresses for workstations or laptops within the organization • They often contain logs on the assignment of IP addresses mapped to the MAC address of the host's NIC → This becomes important if an incident responder has to track down a specific workstation or laptop that was connected to the network at a specific date and time	• Also known as authentication servers, they can retain logs that can show details on successful or unsuccessful logins, credential manipulation, or another credential uses • E.g. we all use a domain controller to login to every service on campus

14

Other Source of Network Evidences



PROXIES

- Organizations can utilize web proxy servers to control how users interact with websites and other internet-based resources
- Web proxies also have additional features such as alerting security personnel to connections to known malware C2 servers or websites that serve up malware.
- A review of web proxy logs in conjunction with a possibly compromised host may identify a source of malicious traffic or a C2 server exerting control over the host

APPLICATIONS

- Application servers also use logs to keep track of access → important in authentication servers
- These servers are connected to the network and often provide some sort of service to the end-user, e.g. Web server, domain authentication server, mail server
- Examples of applications that use logs include Apache, IIS, Microsoft Active Directory server, Microsoft Exchange, etc

15

Network based Artifact



- Digital forensics is evolving from host-centric storage analysis to **network-based evidence** due to increasing online and network attacks
- Network evidence focuses on **devices, traffic flows, and internal routes**, and is critical for detecting:
 - External command-and-control activities
 - Data exfiltration incidents
- Network-based evidence:
 - Complements host evidence through **event corroboration**
 - Improves **root-cause analysis**
- Existing network logs provide an **immediate and valuable evidence source**, supported by proper access knowledge and coordination with IT teams

16

Network Logs / System Logs?

- Network/System logs (i.e. syslogs) are a standard network-based logging protocol that runs on devices connected in a network, allowing them to create (and send) free text-formatted log messages.
- Some logs are kept on device whilst others may be sent to a centralized syslog repository
- Essentially every device on your network whether it's a storage box or a server, a switch or a firewall, likely has a syslog agent you can use to send messages to a common central location.

Common examples of devices using syslogs:

- Operating systems (i.e. Windows with event logs)
- Routers, switches, firewalls, NIDs, DHCP servers, proxies, domain controllers, etc
- Network-based applications

The diagram illustrates a centralized logging architecture. On the left, three categories of devices are shown: 'Workstations' (represented by computer icons), 'Applications' (represented by a monitor icon), and 'Servers' (represented by server rack icons). On the right, two more categories are shown: 'Workstations' and 'Servers'. In the center, a 'Syslog server' is represented by a server rack icon. Arrows from all these device categories point towards the central Syslog server. Below the Syslog server, a box labeled 'Syslog messages' contains an envelope icon, representing the log data being transmitted.

17

Network Logs / System Logs?

- Typical uses of syslogs is system monitoring and management but the same data can also be evidence.

This diagram shows the data flow from a Linux VM to Microsoft Sentinel. A 'Linux VM' box contains a document icon labeled 'Syslog messages'. An arrow labeled 'UDP 514 (Syslog)' points from the VM to a 'Syslog daemon (rsyslog/syslog-ng)' box. From the daemon, two arrows point to an 'Azure Monitor Agent (AMA)' box: one labeled 'TCP 28330 (v. 1.28.11 or higher)' and another labeled 'Unix domain socket (older versions)'. An arrow labeled 'TCP 443 (HTTPS)' points from the AMA to a 'Log Analytics workspace' box, which contains a 'Syslog' icon. The workspace is part of 'Microsoft Sentinel'. Above this diagram, a sample log message is shown with its components: 'Time Stamp' (Oct 10 15:09:23.538), 'Severity' (%LINK-3-UPDOWN), 'Facility' (Interface FastEthernet0/1), 'Mnemonic' (changed state to up), and 'Description' (Interface FastEthernet0/1, changed state to up).

18

Security Information and Event Management (SIEM)

- Network devices have limited log storage, causing logs to be overwritten quickly
- This can result in loss of critical evidence for delayed incident investigations
- A SIEM system addresses this by:
 - Aggregating logs and events from multiple network devices
 - Centralizing data in a single repository
- SIEM enables security teams to:
 - Monitor activity across the entire network
 - Perform faster and more effective incident investigation

SIEM Functions in Incident Response

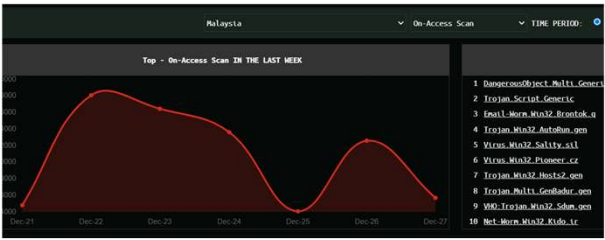
- **Centralized log aggregation** from thousands of network and security devices
- **Long-term log retention** to support compliance and forensic investigations
- **Routine monitoring & reporting** via dashboards and scheduled analysis
- **Real-time alerting & event correlation** to detect suspicious or malicious activity
- **Accelerated incident response** by enabling fast search, analysis, and retrieval of enterprise-wide logs from a single platform

Preparation for Network-Based Evidence Collection

- Effective network evidence acquisition depends on pre-incident preparation
- Lack of preparation can result in missing or delayed critical evidence
- Key organizational requirements:
 - Accurate network documentation
 - Up-to-date network device configurations
 - Centralized log management system

Incident responders must consider legal and regulatory constraints. Clear organizational policies must state:

- System usage is monitored
- Evidence collection is authorized and legally supported



Understanding Network Infrastructure for Forensics



- Effective network forensics requires a **clear understanding of internal network infrastructure**
- Maintain an **up-to-date network diagram** to identify potential evidence sources
- Network diagrams should include:
 - All network components (e.g., routers, switches, wireless access points)
 - Internal IP addressing for rapid remote access during incidents
- Having **standard (baseline) device configurations** enables faster detection of unauthorized changes by attackers

21

NetFlow/Traffic Flows



- NetFlow is a feature found in Cisco™ network devices such as switches and routers that allows network administrators to monitor traffic within the network → reporting is done via UDP messaging to a centralized collector
- While not strictly a security tool, but it does provide a good deal of data to incident responders in the event of an incident
- Participating devices forward the NetFlow data to the collector, which includes the IP addresses, protocols, and data size.
- This data is critical to providing incident response analysts with details that they may not normally otherwise acquire
- Configuring NetFlow is dependent on the type and manufacturer of the network components
- Not every organization may have the budget/requirements to sustain NetFlow logging

22

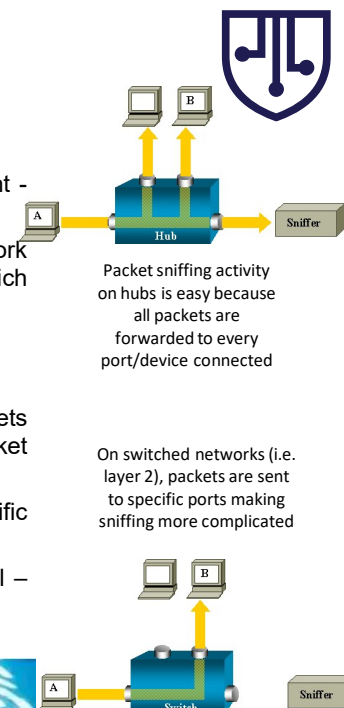
NETWORK PACKET CAPTURE & ANALYSIS



23

Packet Captures

- Capturing network traffic is critical to having a full understanding of an incident - an alternative to NetFlow is a packet capture or network tapping
- A network tap is a system (sometimes called a sniffer, packet analyzer, network analyzer) that is in line with the compromised host and the switch which eavesdrops on the communications between parties
- Where the system is located will indicate what traffic is captured
 - Could be entire segments of the network or specific devices
- On networks connected with hubs, packet capture is easy because the packets are sent to everyone connected and the hosts themselves determine which packet to retain/drop
- On switched networks, it gets tricky because packets are only delivered to specific hosts on the network (via routing table)
- There are many applications available, both free/open-source and commercial – that can perform capture operations



24



Network Packet Analysis

Network packets

- Data sent across the network is divided into chunks called packets
- Packets are divided into three sections: header, payload, footer

Packet headers

- Can have several headers put on by different protocols at different layers of the OSI model
- Each header contains different information; combined, they have several pieces of information interesting to forensic investigators
- TCP header synchronization bits also yield interesting forensic data
- Port scanning: seeing incoming packets destined for well-known ports is a telltale sign of a port scan

25



Network Packet Analysis (Cont.)

Payload

- The body or information content of a packet
- Actual content that the packet is delivering to the destination
- If packet is fixed length, payload may be padded with blank information or a specific pattern to make it the right size

Trailer

- Contains error-checking data to detect errors that occur during transmission
- May be part of the Ethernet or Point-to-Point Protocol (PPP) frame or other Layer 2 protocol

Ports

- A number that identifies a channel in which communication can occur
- 65,535 possible ports
- Knowing what port a packet was destined for (or coming from) tells you what protocol it was using

26

Some Important Ports & Intruder Ports



Port Number	Protocol Using That Port
20, 21	File Transfer Protocol (FTP)
22	Secure Shell (SSH)
23	Telnet
25	Simple Mail Transfer Protocol (SMTP)
53	Domain Name Service (DNS)
80	Hypertext Transfer Protocol (HTTP)
110	Post Office Protocol Version 3 (POP3)

Common/ Important Ports

Port Number	Utility
407	Timbuktu
6666	Beast
23476/23477	Donald Dick
31337	Back Orifice
43188	Reachout
54320/54321	B02K

Common Intruder Ports



27

Network Attacks



Denial of service (DoS)	Ping of Death	Ping flood	Teardrop
SYN flood	Land attack	Smurf	DHCP starvation
HTTP post attack	PDoS	Login DoS	Packet mistreating

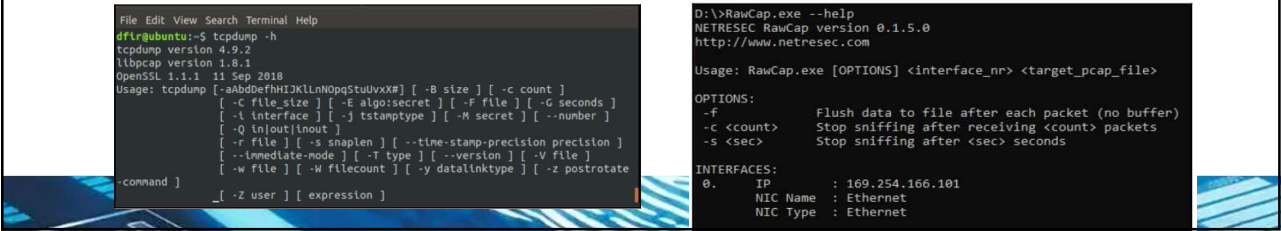


28



Network Traffic Analysis Tools: Sniffer

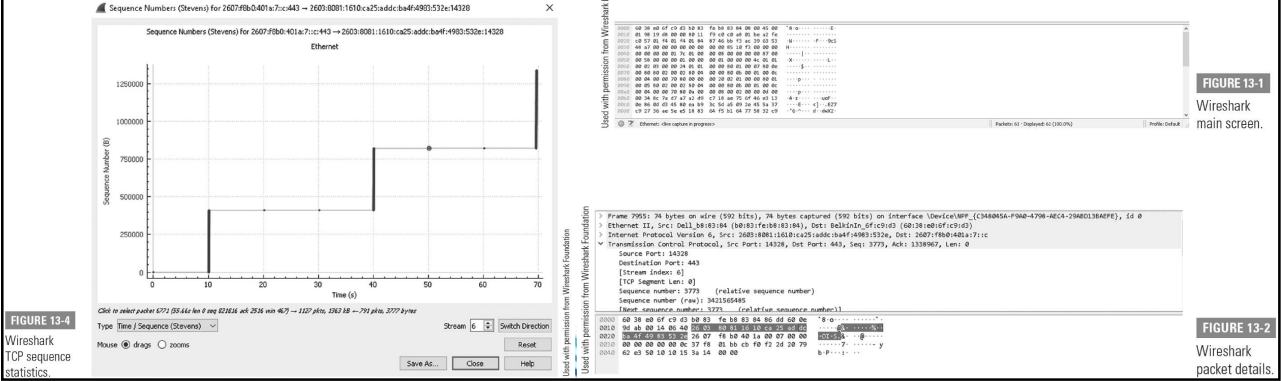
- Software or hardware that can intercept and log traffic passing over a digital network
- Extracts network packets and performs a statistical analysis on the dumped information
- Commonly applied sniffers include **Tcpdump** (Unix platforms and) **WinDump**
 - Tcpdump
 - A commonly applied sniffer for various UNIX platforms
 - WinDump
 - A version of Tcpdump for Windows
- Both programs extract network packets and perform statistical analysis on dumped information



29

Wireshark

- Packet sniffer and network protocol analyzer
- Analyzes real-time and saved data
- Supports IPv4 and IPv6
- Freely available



30



Nmap

- Allows the user to map out what ports are open on a target system and what services are running
- Is a command-line tool but has a Windows interface (Zenmap) that is available for free
- Is popular with hackers because it can be configured to operate rather stealthily and determine all open ports on an individual machine, or for all machines in an entire range of IP addresses
- Is popular with administrators because of its ability to discover open ports on the network
 - Could indicate spyware, a backdoor, or many other attacks

```
+ -- sudo nmap -sS scanme.nmap.org
Starting Nmap 7.93 ( https://nmap.org ) at 2022-11-16 12:59 EST
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.081s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
135/tcp   filtered mcrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
593/tcp   filtered http-rpc-epmap
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 2.64 seconds
```

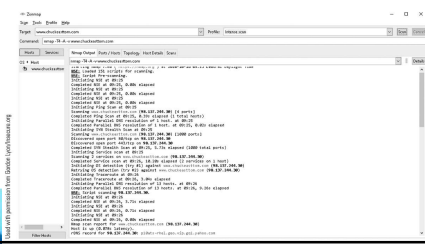


FIGURE 13-5
Nmap

31



Snort and NetWitness

Snort

- Primarily used as an open-source intrusion detection system
- Can function as a robust packet sniffer with a lot of configuration options

NetWitness

- Product from RSA that has a freeware version
- Threat analysis/protocol analyzer
- Not as easy to use as Wireshark

32



Other Tools

Commercial

- SolarWinds DPI Network tool
- ManageEngine NetFlow Analyzer
- Omnippeek Network Protocol Analyzer
- Fiddler
- Commercial with free/limited version
- Paessler Packet Capture Tool
- Capsa

Open source / Free

- tshark – CLI-based version of WireShark
- NetworkMiner
- Arkime

33

WIRELESS NETWORK



34

Forensic Concerns with Wireless Networks

Did a perpetrator use a wireless network entry point for a direct network attack or theft of data?

Did an attacker use a third-party wireless network, such as a hotel hotspot, to conceal his or her identity?

35

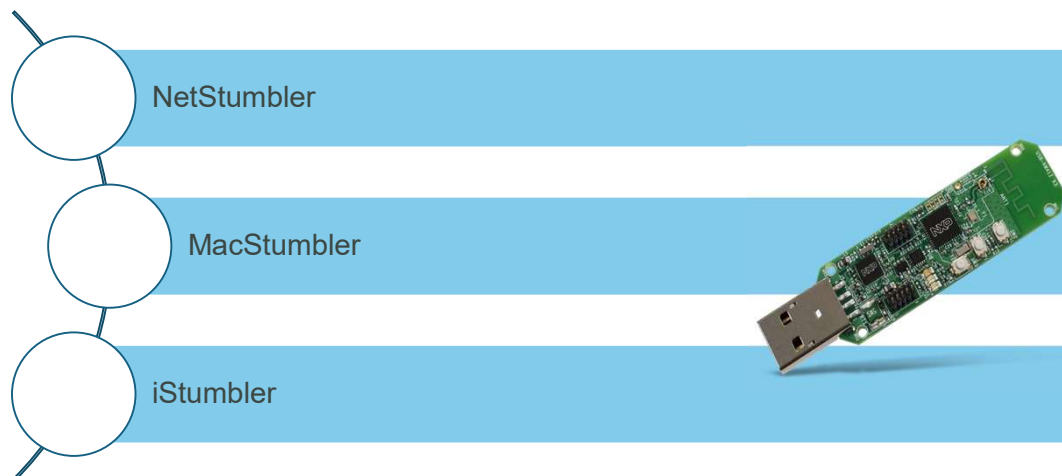
Wireless Storage Devices

- Wireless digital and video cameras
- Wireless printers with storage capacity
- Wireless network-attached storage (NAS) devices
- Tablets and smartphones
- Wireless digital video recorders (DVRs)
- Wireless game consoles

36



Wireless Network Discovery Tools



37



Network Evidence Collection

- In order to conduct a proper examination of log files and other network data such as packet captures, they often have to be moved from the log source and examined offline (i.e. detailed analysis in lab)
- Log files or packet captures have to be handled with due care to ensure that they are not corrupted or modified during the transfer
- The acquisition of network evidence such as a packet capture or log file should be thoroughly documented and verifiable (for CoC purposes)
- Forensic practitioners should ensure that they can trace back every separate piece of evidence to its source as well as the date and time that the evidence was collected
- This can be recorded in a network evidence log sheet and entries completed for each piece of evidence

38



Network Evidence Collection (Cont.)

- A possible log entry captures the following information:
 - File Name: Each log file or packet capture should have its own unique name.
 - Description: A brief description of the file.
 - Location: The location is important as there could be more than one source for network capture
 - Date and Time: When file was transferred to the medium.
 - Collected By: Initials are sufficient for the log file but follow SOP
 - MD5 Hash: Fingerprint the file with a verifiable signature

File Name	Description	Location	Date	Time	Collected By	MD5 Hash
Ping_capture	Packet Capture of Ping activity	192.168.2.1	6/26/19	1642	GTJ	7e559dc8eeeb66115566d93f96e7dfb8

39



Summary

What was covered today:

- There is a wealth of information available from network devices spread throughout the environment and a forensic practitioner should have the knowledge and tools to extract evidence from it
- The different points on a network means different tools (software/hardware) are necessary to enable proper evidence capture



40

